



REGIONAL SALES ANALYSIS

Exploratory Data Analysis

PROBLEM STATEMENT & OBJECTIVE

- Current sales strategies lack data-driven alignment with revenue and profit drivers across regions, channels, and products, leading to suboptimal performance and high concentration risk.
- Utilize EDA to **identify core drivers**, **uncover trends/outliers**, and **measure performance** to strategically optimize pricing, promotions, and expansion for sustained growth and reduced risk.

APPROACH

- **Exploratory Data Analysis (EDA):** Clean, transform, and visualize sales and profit data.
- **Driver Identification:** Use decomposition (e.g., Pareto analysis) to pinpoint the products, channels, and regions contributing most to revenue and profit.
- **Temporal Analysis:** Apply time-series plots and decomposition to uncover seasonality, long-term trends, and sales outliers.
- **Strategic Recommendations:** Translate findings into actionable recommendations for optimizing pricing, improving promotional effectiveness, and identifying low-risk expansion opportunities.

DATA OVERVIEW

Category	Key Metrics/Dimensions	Data Snapshot
Source & Scope	Source:	Corporate Sales & Budget Datasets
	Sales Time Period:	4 Years (January 1, 2014, to December 31, 2017)
	Budget Scope:	2017 Product Budgets only
Key Metrics	Total Revenue:	\$1.24 Billion (approx.)
	Total Gross Profit:	\$461.8 Million (approx.)
	Total Records:	64,104 individual sales transactions
Grouping Variables	Products:	30 unique products
	Sales Channels:	3 major channels (Wholesale, Distributor, Export)
	Regions:	993 unique delivery region indices (requires mapping/joining to Regions and State Regions tables)

PROJECT WORKFLOW

- **Data Consolidation & Preparation**
 - **Acquisition & Merge:** Loaded all six CSV sheets and created a **single master data table** by joining them (Customers, Products, Regions, Budgets) to the Sales Orders base.
 - **Cleansing & Scope:** Filtered data to ensure a clean, four-year analysis period (**2014–2017**) and removed extraneous geographical and order-related columns.
- **Metric Generation & Dimensional Enrichment**
 - **Financial Metrics:** Calculated the primary performance indicators: **Profit** (Revenue - Total Cost) and **Total Cost Per Transaction**.
 - **Temporal Dimensions:** Derived Year and segmented orders into four calendar **Seasons** (Winter, Spring, Summer, Autumn) from the OrderDate.

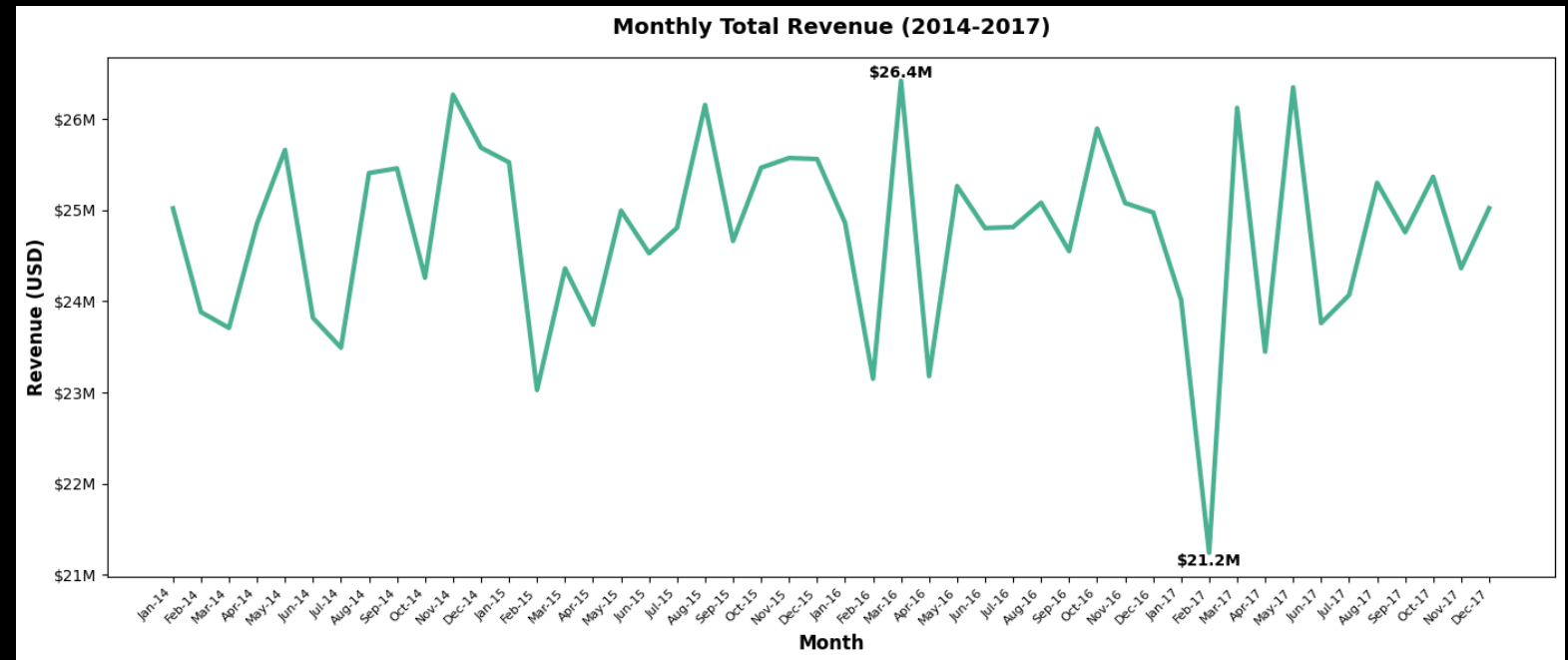
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

Annual Seasonality:

Revenue typically peaks in the **Summer/Autumn** months and drops slightly in **Winter**, suggesting a recurring seasonal sales cycle.

Monthly Revenue Trend (2014–2017)



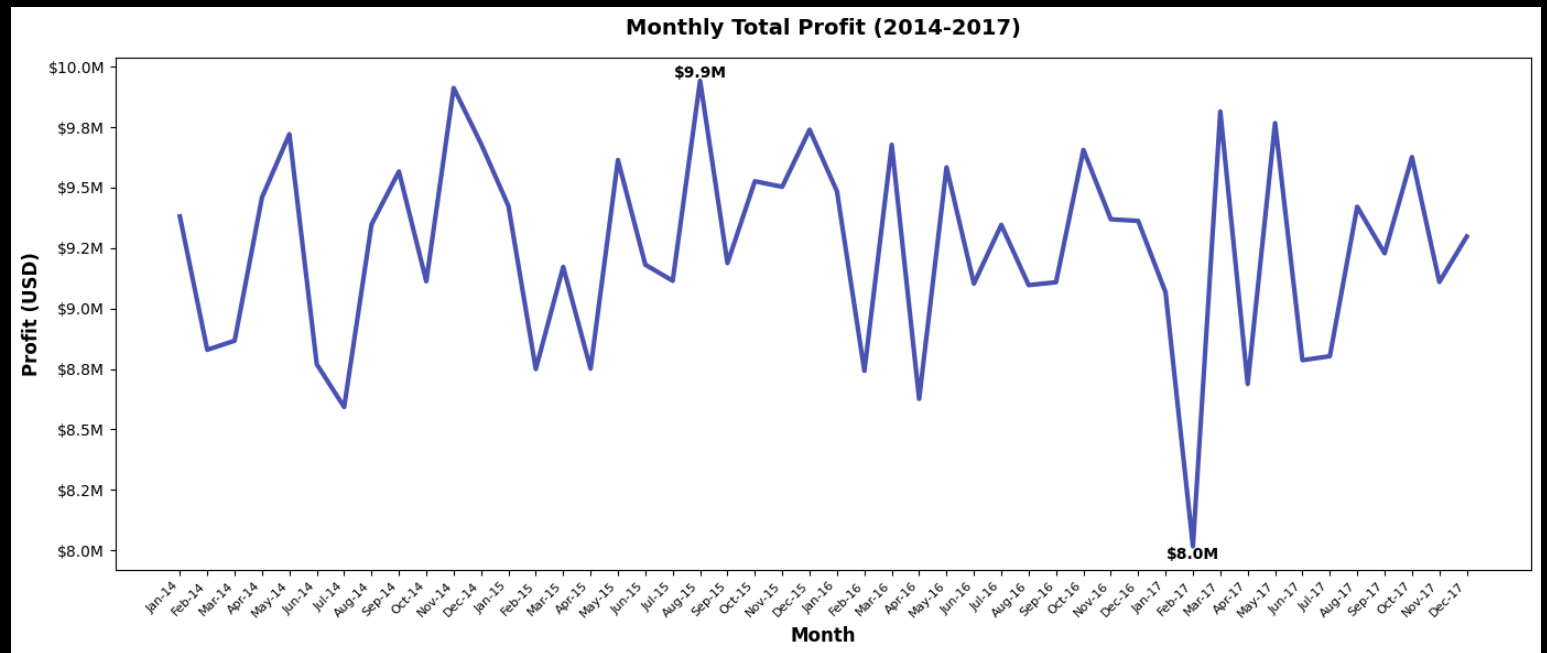
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

Profit Mirrors Revenue:

The monthly profit curve closely follows the revenue curve, suggesting **stable cost of goods sold** relative to sales prices over the 4-year period.

Monthly Profit Trend (2014–2017)



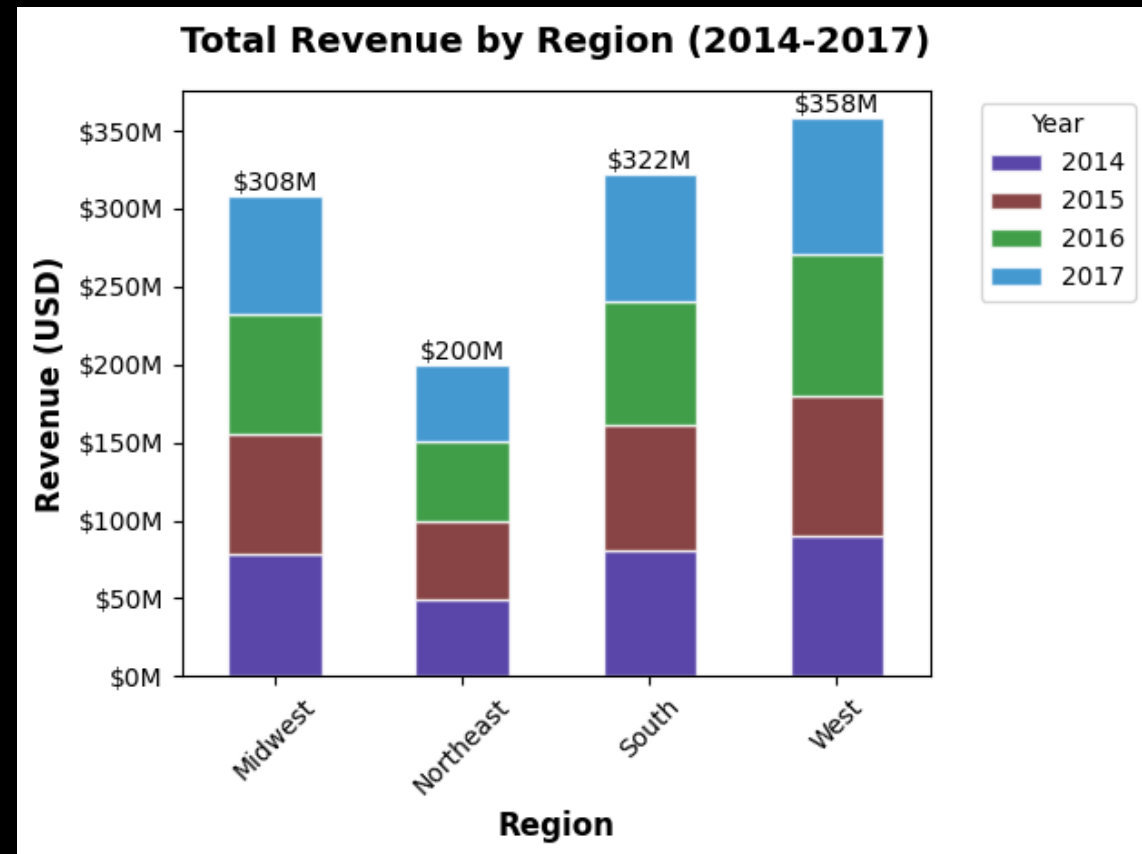
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

West Dominance:

The **West** region is the unequivocal revenue leader, consistently contributing the largest total sales volume across all four years.

Regional Revenue Performance (YOY)



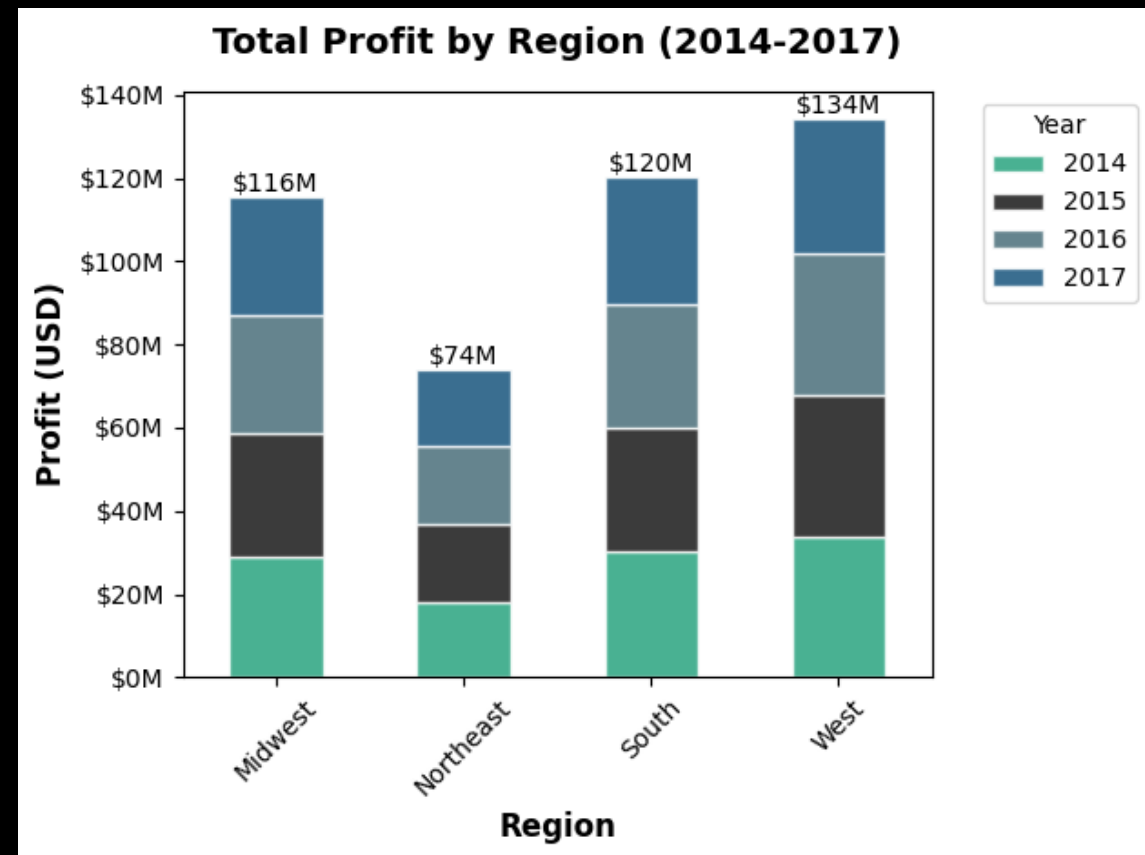
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

Margin Consistency:

Comparison with the Revenue chart shows all regions maintain similar profit margins over time, with no single region demonstrating unusually low or high profitability.

Regional Profit Performance (YOY)



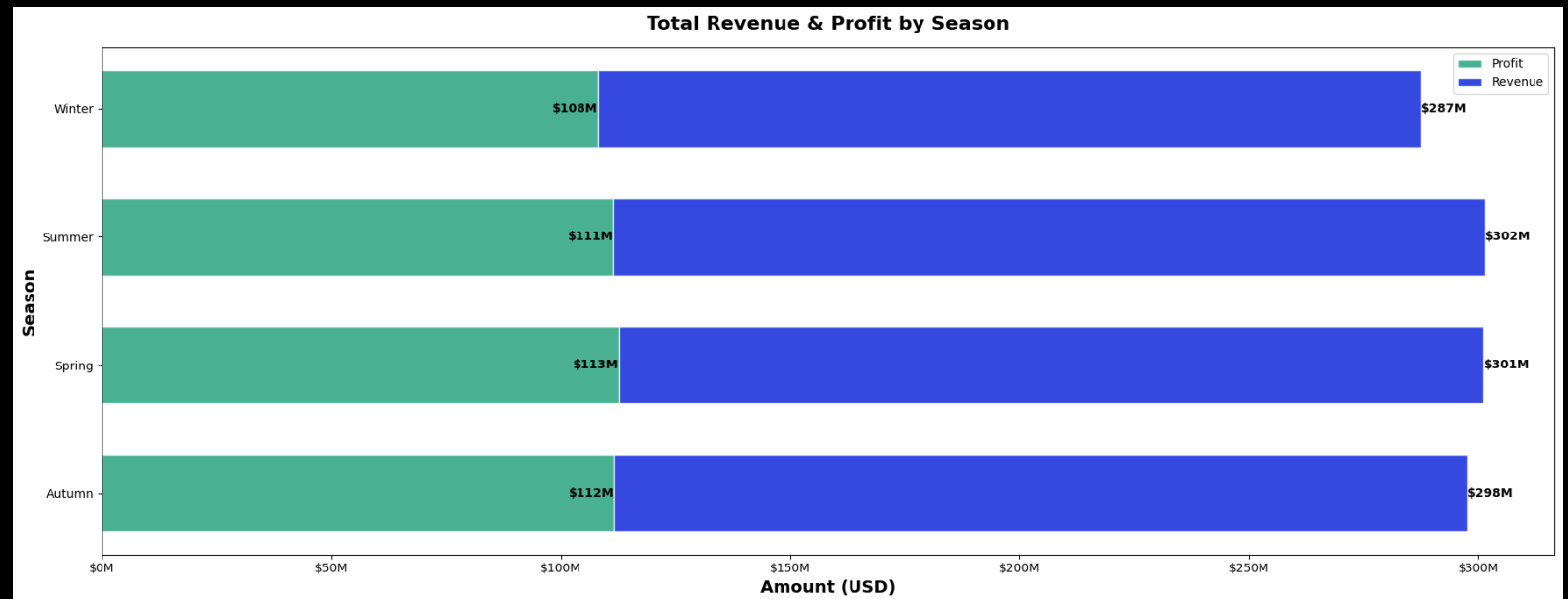
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

Lowest Performance:

Winter consistently performs the lowest in both total Revenue and total Profit, confirming the annual seasonal slowdown identified in the monthly analysis.

Seasonal Financial Summary



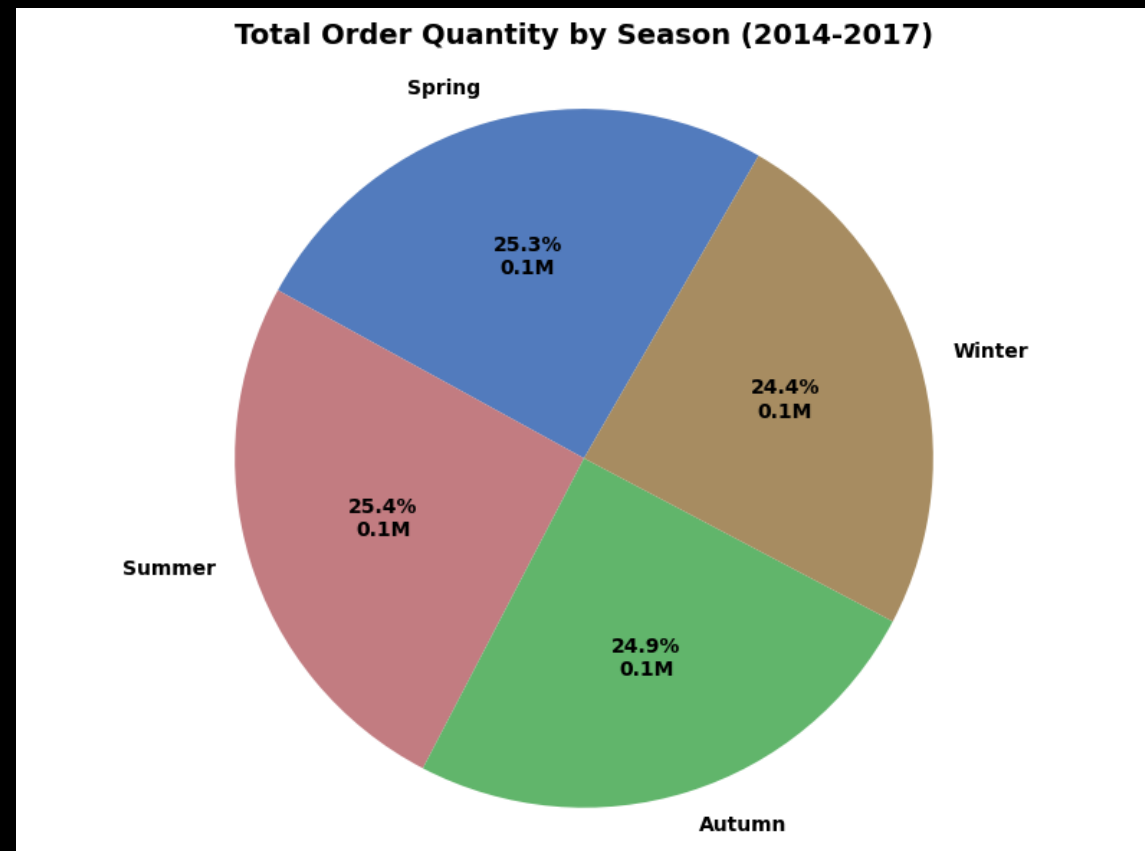
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

Volume vs. Value:

The difference between this chart and the Seasonal Financial Summary chart suggests that higher-value items are more frequently sold in **Spring/Autumn**, while lower-value items may compensate in volume during other seasons.

Order Quantity by Season



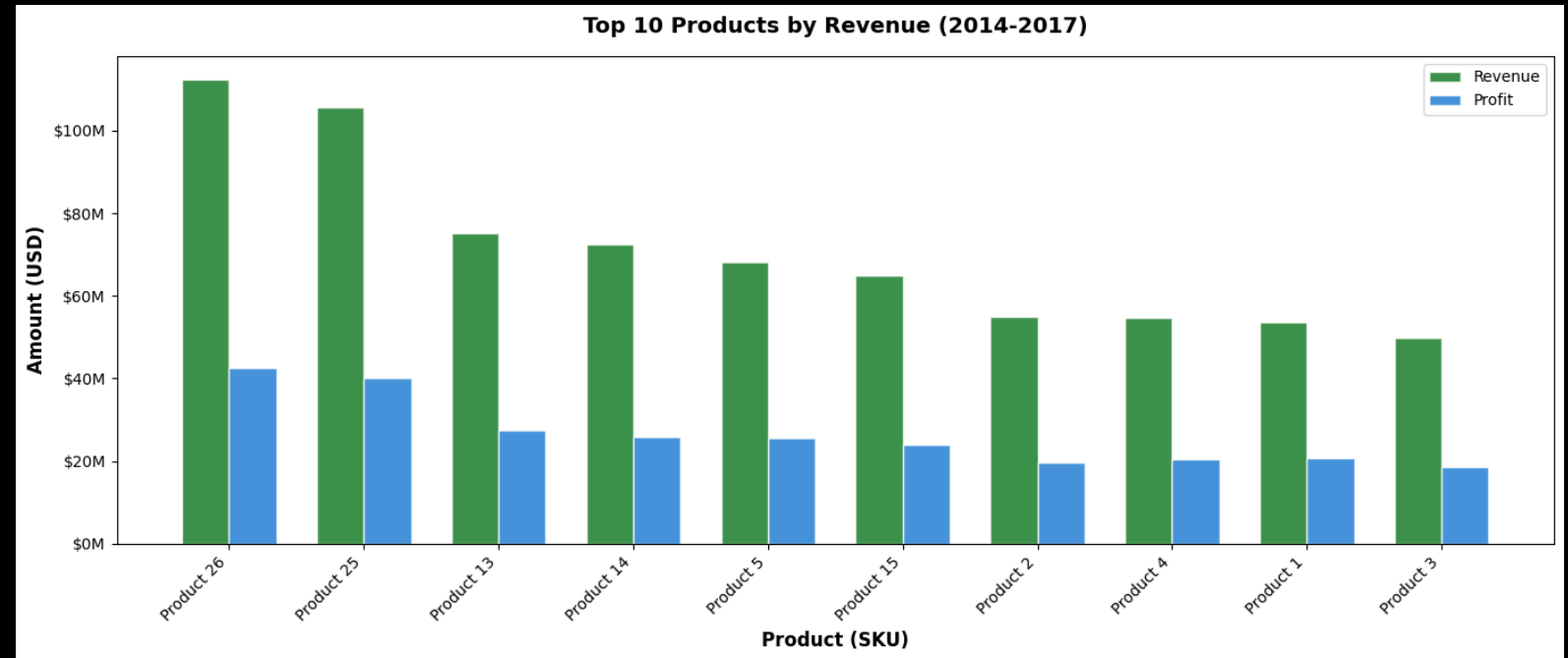
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

Product Purity:

Most top-revenue products are also top-profit products, signifying strong gross margins across the high-volume SKU mix.

Top 10 Products: Revenue vs. Profit



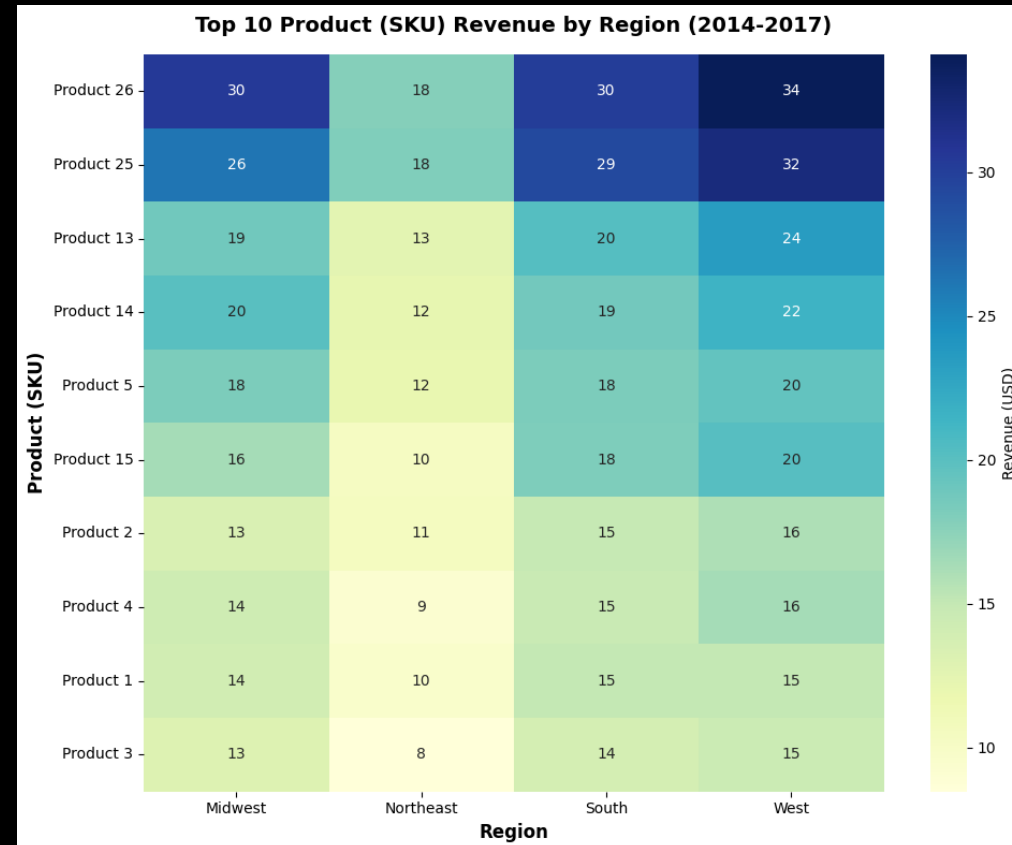
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

West & South Concentration:

Product sales are heavily concentrated in the two largest regions, **West** and **South**.

Top 10 Product Revenues by Region



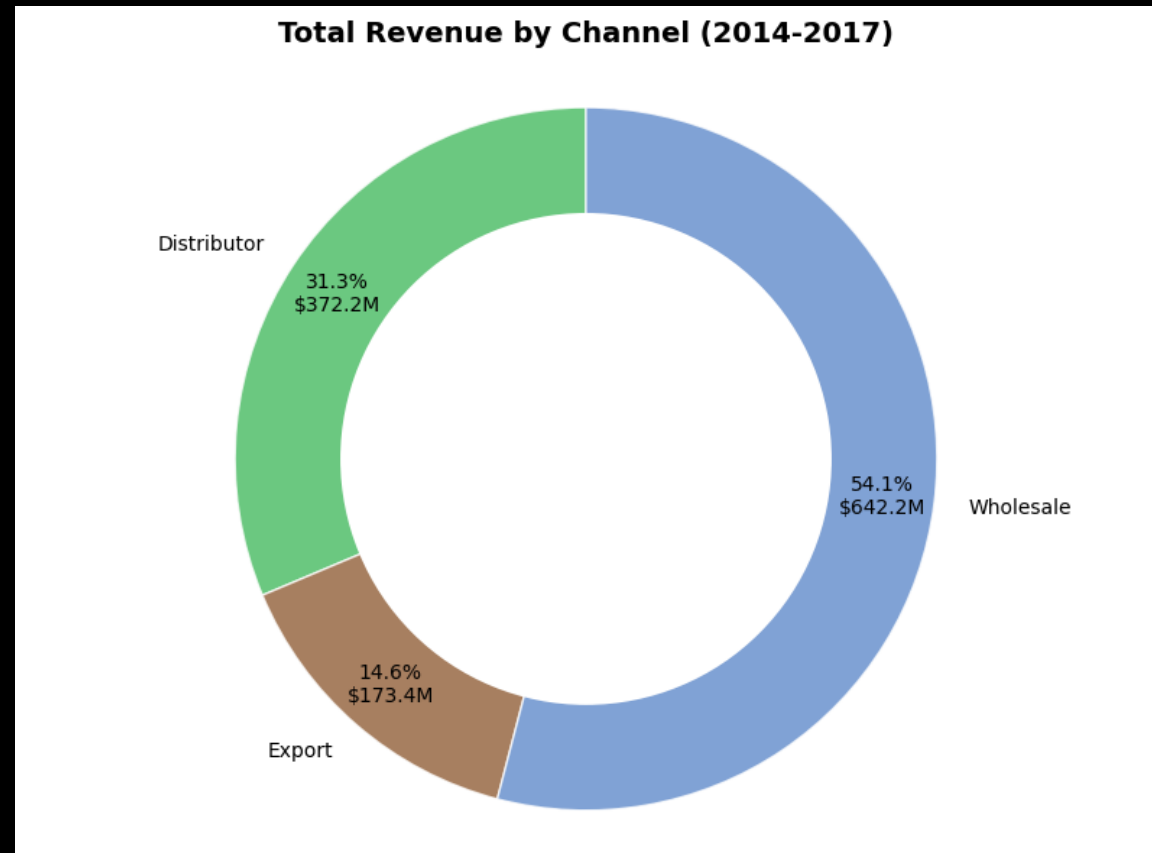
EXPLORATORY DATA ANALYSIS (EDA)

Key Insight

Dominant Channel:

Identified **Wholesale** as the single channel that delivers the largest percentage of total sales revenue.

Total Revenue by Channel



RECOMMENDATIONS

- **Expand Geographically:** The **West and South regions dominate** revenue and profit. Target the **Midwest** and **Northeast** for market expansion using the **Top 10 SKUs** to diversify risk and capture untapped potential.
- **Mitigate Winter Slump:** Implement a focused **Q1 Revenue Protection Strategy** (e.g., promotional bundles or incentives) to offset the recurring financial and volume drop seen during the **Winter season**.
- **Prioritize Top-Tier SKUs:** Dedicate the majority of inventory, production, and marketing spend to the **Top 10 Products**. Their performance drives a disproportionate share of total profit and requires focused resource allocation.
- **Optimize Channel Margin:** Review the cost-to-serve for the highest revenue-generating channels. **Audit the largest channels** to ensure their **Profit Share** is proportional to their **Revenue Share**, flagging any that operate with surprisingly low margins.